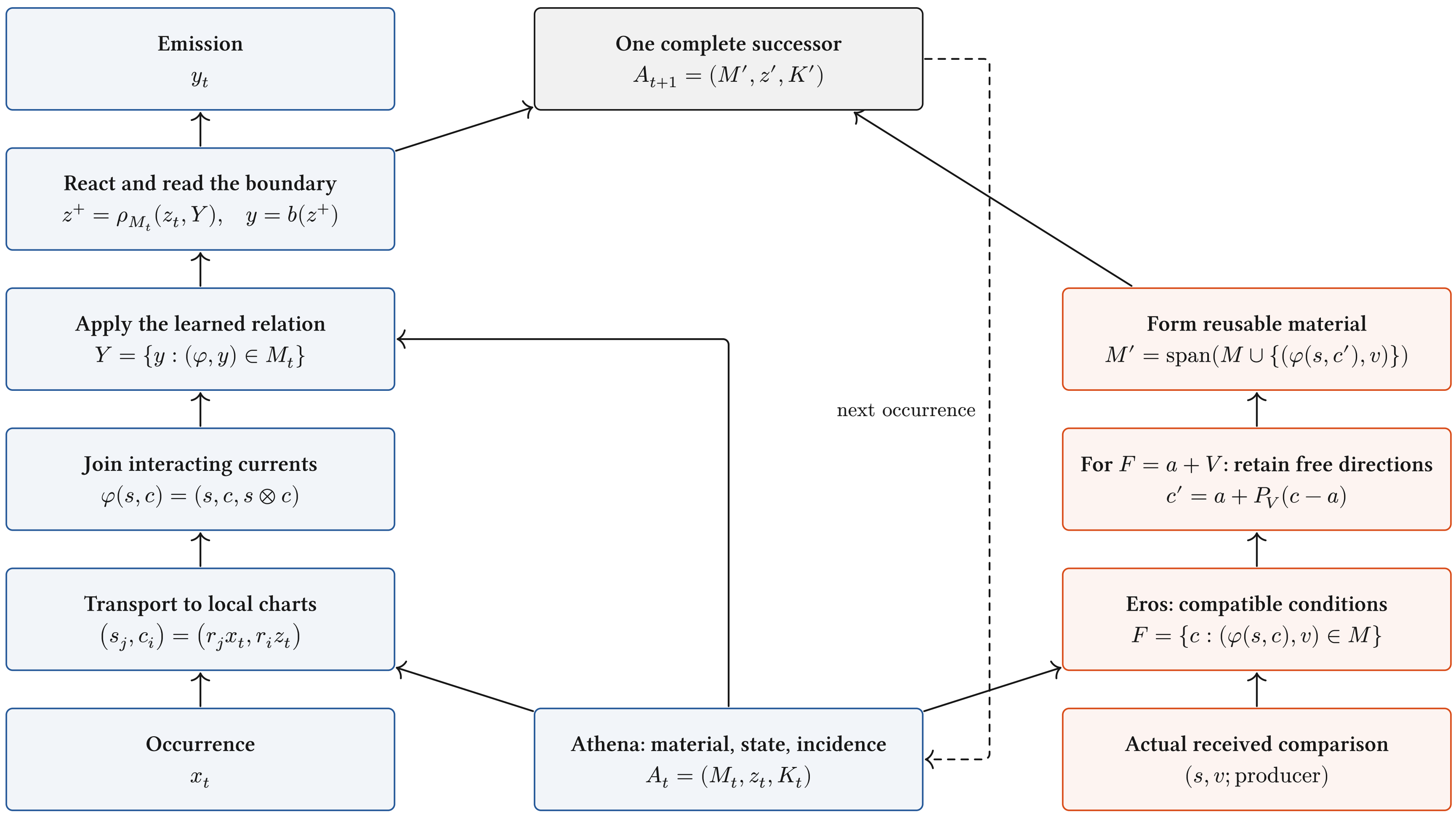
PLATE 01

Athena conducts. Eros develops.

One continuing body; explicit current paths and a joined developmental return.



Definition / source-inspected binding. Blue: conduct. Red: Eros's local relation/condition update. Source, material and the comparison retain their actual producing cut.

Interpretation: full assembly. Local passages compose over admitted incidence. The bilinear relation above is an existing binding; complete contextual assembly remains in progress.

PLATE 02

What a layer does to a space

Finite vector charts: the operation, its categorical role, and the distinctions it retains.

Linear map / "projection"

$$x \mapsto Wx$$

$$V \xrightarrow{W} \operatorname{im} W$$

Linear morphism. Fibres are $x + \ker W$; rank controls the image. It is a projection only when $W^2 = W$ on one space.

Concatenate / retain both

$$(x, y) \mapsto x \oplus y$$

$$\begin{tikzcd} V \arrow[dr, "i_V"] & \\ & V \oplus U \\ U \arrow[ur, "i_U"] & \end{tikzcd}$$

Biproduct injections retain both components. Combining their storage does not yet identify them.

Add / superpose

$$(x, y) \mapsto x + y$$

$$V \oplus V \xrightarrow{\nabla} V$$

The codiagonal identifies $(x + h, y - h)$. Opposed contributions can cancel at this receiver.

Tensor / interact

$$(x, y) \mapsto x \otimes y$$

$$V \times U \xrightarrow{\otimes} V \otimes U$$

Universal bilinear interaction. A bilinear response factors through a linear map out of $V \otimes U$. Gating adds a declared nonlinearity.

Softmax / relative participation

$$p_i = \frac{e^{s_i}}{\sum_j e^{s_j}}$$

$$\mathbb{R}^n / \langle 1 \rangle \xrightarrow{\tilde{\sigma}} \operatorname{int} \Delta^{n-1}$$

Smooth bijection of the score-difference chart with the simplex interior. All $p_i > 0$; common score shifts disappear.

Normalize / choose a section

$$N_0(x) = \sqrt{d}\frac{x}{\|x\|}$$

$$\big(\mathbb{R}^d \setminus \{0\}\big)/\mathbb{R}_{>0} \longrightarrow S^{d-1}_{\sqrt{d}}$$

Zero-offset RMS normalization chooses one point per positive ray. Layer centering additionally removes the common-value direction.

$$N_\varepsilon(x) = \frac{x}{\sqrt{\frac{\|x\|^2}{d} + \varepsilon}}, \quad \varepsilon > 0 : \quad \mathbb{R}^d \simeq \left\{y : \|y\| < \sqrt{d}\right\}$$

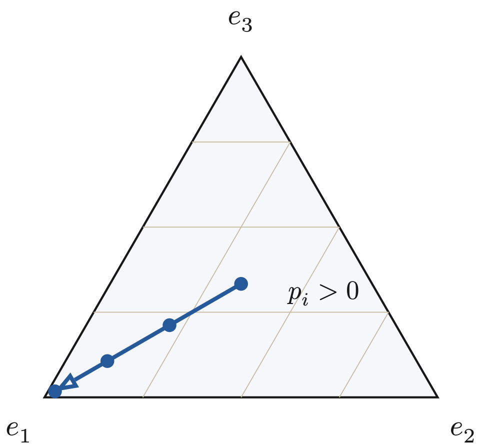
Proved-standard, stated finite charts. Positive ε retains radial scale before learned gains: stabilized RMS normalization is not the exact ray quotient above. Zero gain coordinates can erase further information.

PLATE 03

Smooth redistribution, selection walls, rank changes

A geometric distinction between moving within a chart and changing the active continuation.

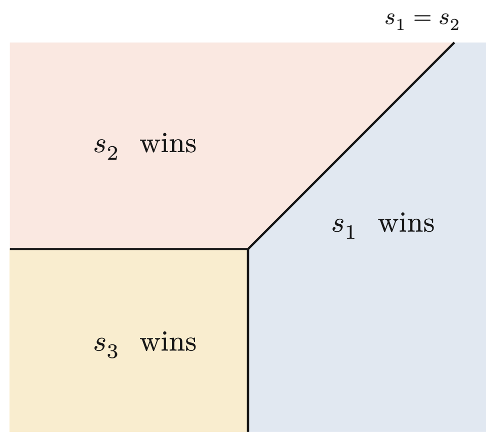
Softmax stays inside the simplex



$$s = (\lambda, 0, 0), \quad p = \frac{1}{\exp(\lambda)+2}(\exp(\lambda), 1, 1)$$

Every finite λ keeps three positive shares. The blue path approaches a vertex only in the limit.

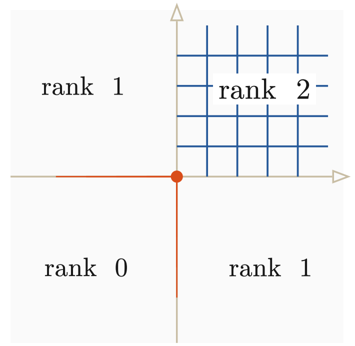
Hard selection crosses a wall



$$E(s) = \operatorname{argmax}_i s_i, \quad s = (a, b, 0)$$

The maximal-score tie walls divide parameter space into regions with different active indices. A tie convention is part of the map.

A nonlinear map changes local rank



$$r(x)_i = \max(0, x_i), \quad Dr = \operatorname{diag}\big(1_{x_i>0}\big)$$

The four sign chambers form a support lattice. Whole negative directions collapse; the derivative changes at its walls.

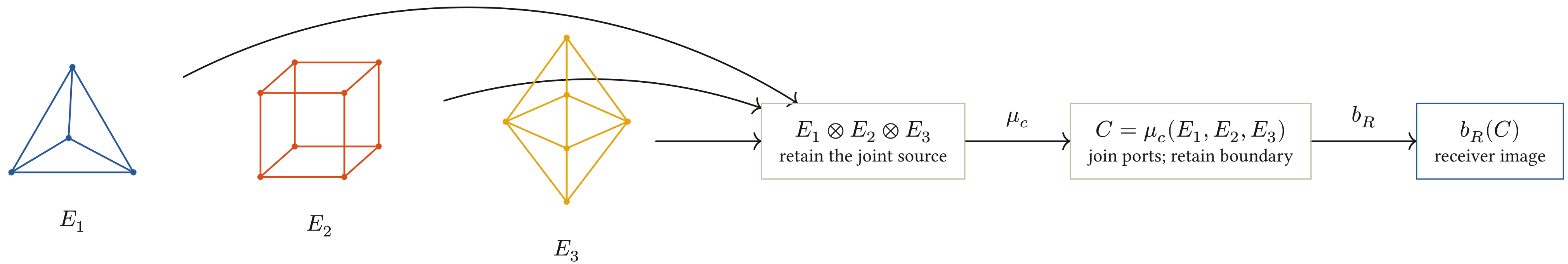
$$\begin{tikzcd} & \{1\} \arrow[dr] & \\ \varnothing \arrow[ur] \arrow[dr] & & \{1,2\} \\ & \{2\} \arrow[ur] & \end{tikzcd}$$

$$M_t = \begin{pmatrix} 1 & 0 \\ 0 & t \end{pmatrix} \longrightarrow \operatorname{rank} M_t = \begin{cases} 2 \text{ if } & t \neq 0 \\ 1 \text{ if } & t = 0 \end{cases}$$

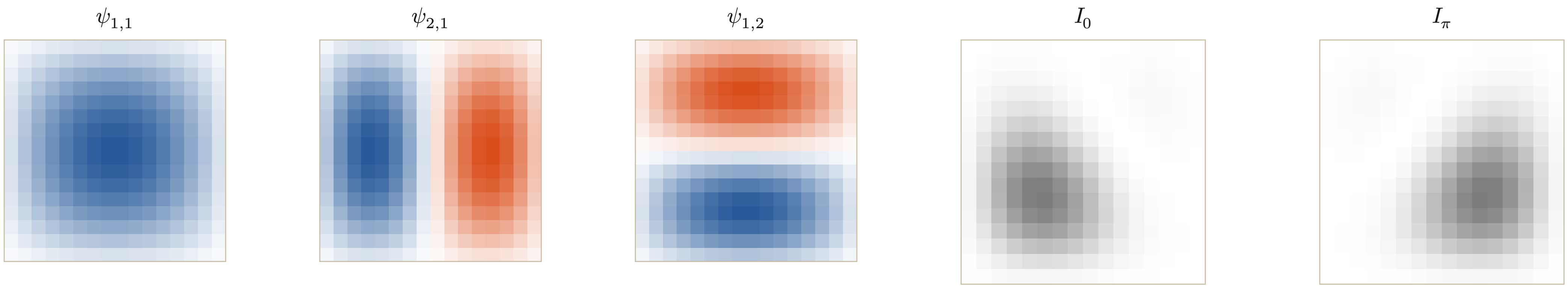
Proved-derived finite charts; interpretation of phase. Smooth participation, active-set change and rank loss are different events. Information Chemistry calls a change of the continuing transport class a phase transition; a physical phase claim additionally requires its constitutive law.

Information Chemistry: elements into compounds

Typed ports compose; relative phase changes the receiver image of the same constituent modes.



Interpretation. Polyhedra depict ported boundary cells, not semantic labels. The lattice tiles below give a separate explicit realization of constituent composition.



$$\psi_{m,n}(i,j) = \sin\left(\frac{m\pi i}{17}\right)\sin\left(\frac{n\pi j}{17}\right), \quad i,j \in \{1,\ldots,16\}$$

$$u_\theta = \psi_{1,1} + 0.7e^{i\theta}\psi_{2,1} + 0.5\psi_{1,2}, \quad I_\theta = |u_\theta|^2$$

$$|a+b|^2 = |a|^2 + |b|^2 + 2\,\mathrm{Re}\left(a\bar{b}\right)$$

Definition / computational illustration. Dirichlet square-lattice modes, dimensionless; shared intensity scale 0 to 4.84. Blue/red mark positive/negative amplitude. Interference changes the output without changing the three modes.

Conjecture to develop: one reusable family of typed, parameterized generators can compose every admitted behavioral class. A pattern is a whole diagram modulo its declared future receivers, not a unique solid or a stored output image.

PLATE 05

When diagrams agree, Soulkiller can lift their maps

Compare complete transitions; then restrict at an explicitly declared receiver family.

A commuting architecture

$$
\begin{tikzcd}
A \times X_A \arrow[r, "F_A"] \arrow[d, "q_{\mathrm{in}}"] & Y_A \times T_A \times A \arrow[d, "q_{\mathrm{out}}"'] \\
B \times X_B \arrow[r, "F_B"] & Y_B \times T_B \times B
\end{tikzcd}
$$

$q_{\mathrm{out}} F_A = F_B q_{\mathrm{in}}$

Invertible maps: re-expression.
Noninvertible maps: retained fibres.
Unequal paths: explicit defect.

$X \leftarrow W_f \rightarrow Y \leftarrow W_g \rightarrow Z$

$W_{g \circ f} = W_f \times_Y W_g$

Definition / proved-derived composition. Serial joining retains the actual occurrence pullback. The same state map appears before and after each transition.

Soulkiller: realization, then restriction

$$
\begin{tikzcd}
\text{Source diagram } D_S \arrow[rr, "i"] & & \text{Native realization } D_N \arrow[rr, "r"] \arrow[d] & & \text{Retained body } D_R \\
& & \text{Excite and intervene (actual receiver/history family)} \arrow[urr] \arrow[dll] \arrow[d] \arrow[drr] & & \\
\text{Cold witness} & & \text{Native material} \arrow[d] & & \text{Insufficiency} \\
& & \text{Eros} \rightarrow \text{successor Athena} & &
\end{tikzcd}
$$

$F_N i = i F_S, \quad r F_N = F_R r$

$\therefore F_R(ri) = (ri)F_S$

Conditional lift. The equations state the required scope. Existing SKE evidence covers its declared families; a full V4.1 binding remains open.

Development also owes: $q(U_A(M, v)) = U_B(q(M), v)$

Conditional. Equal inference maps need not have equal learning updates. State, shared parameters, producing factors and the update metric travel with the developmental square.

Sources: **Categorical Holonics**; **Elements of Holonics**; the July 19 Information Chemistry record; the August 9 crystal/receiver record; current HNN composition and source owners.
Diagrams: Fletcher 0.5.8, with the repository's geometric notation and CeTZ. Mathematical interpretations and conjecture do not change implementation grades.